

Disconnected spectra for constrained Toeplitz operators

Candidate counterexamples and an essential-spectrum theorem
for Question 6.1 of arXiv:2110.06379

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Answer to two parts of Question 6.1. Ordinary spectral connectedness is false in both the Neil and two-point settings, even for the bounded analytic rational symbol

$$\phi(z) = \frac{z}{(1 - z/2)^2}.$$

In contrast, for every L^∞ symbol and every finite-codimensional Hardy subspace, the constrained compression has exactly the same Fredholm essential spectrum as the classical Toeplitz operator. Its essential spectrum is therefore always connected.

1 The source question

Felder, Pfeffer, and Russo [1] consider compressions

$$T_\phi^M = P_M(\phi \cdot)|_M$$

to representation-carrying finite-codimensional subspaces $M \subset H^2$. Their Question 6.1, on source PDF page 17, asks whether the spectrum, essential spectrum, and relative point spectrum are connected for arbitrary $\phi \in L^\infty$ in the two-point and Neil settings. Their Question 6.2 asks the relative-spectrum question for general finite-codimensional subalgebras.

2 Idea of the proof

Each Neil or two-point Hardy space in the examples splits as

$$M = \mathbb{C}f \oplus \Theta H^2,$$

where Θ is a degree-two finite Blaschke product. An analytic symbol preserves ΘH^2 . Relative to this splitting, its constrained Toeplitz operator is lower triangular:

$$T_\phi^M \cong \begin{pmatrix} \gamma & 0 \\ * & M_\phi \end{pmatrix}, \quad \gamma = \frac{\langle \phi f, f \rangle}{\|f\|^2}.$$

Thus any γ outside $\phi(\overline{\mathbb{D}})$ becomes an isolated spectral point. The chosen rational ϕ and explicit f make this happen in both settings.

The essential-spectrum statement is even simpler. Extend the compression by zero on the finite-dimensional orthogonal complement of M . The extended operator differs from the classical T_ϕ by finite rank. Fredholm essential spectra are invariant under that operation and under adding a finite-dimensional direct summand.

Again, the behavior of ϕ is quite restricted. All told, in order for a constrained Toeplitz operator to have non-empty relative spectrum in the real-valued setting, it must have a symbol (up to a translation) with set of positivity (negativity) coinciding with that of $\operatorname{Re}(c(k_a - k_b))$ for some $c \in \mathbb{C}$.

Reflecting back to Section 1 Widom's connected-spectrum result held for arbitrary symbols. A result of this generality is still unknown for the two-point and Neil cases:

Question 6.1 (Open). *For general $\phi \in L^\infty$, are the spectrum, essential spectrum, and relative point spectrum of T_ϕ^t connected? In the Neil algebra setting?*

The Neil and two-point constraints are, heuristically speaking, the building blocks for general finite codimensional subalgebras of H^∞ . This notion, due to Gamelin, was formalized in [21, Theorem 9.8], and reinterpreted in [27, Theorem 2.1]. A natural generalization of the work here, and in [11], would be to consider spectra of Toeplitz operators associated to these algebras:

Question 6.2 (Open). *For a general finite codimensional subalgebra of H^∞ , do the associated Toeplitz operators have connected relative spectrum?*

Here, by ‘associated’ we mean Toeplitz operators acting on subspaces of H^2 which carry a representation for the subalgebra.

Figure 1: Questions 6.1 and 6.2 in the source PDF, page 17.

3 A triangular compression lemma

Lemma 1. *Let Θ be a finite Blaschke product, let $0 \neq f \in K_\Theta := H^2 \ominus \Theta H^2$, and suppose $M = \mathbb{C}f \oplus \Theta H^2$ orthogonally. If ϕ is analytic on a neighborhood of $\overline{\mathbb{D}}$, put*

$$\gamma_f = \frac{\langle \phi f, f \rangle}{\|f\|^2}.$$

If $\gamma_f \notin \phi(\overline{\mathbb{D}})$, then

$$\sigma(T_\phi^M) = \{\gamma_f\} \cup \phi(\overline{\mathbb{D}}).$$

In particular, this spectrum is disconnected and γ_f is an isolated eigenvalue.

Proof. Multiplication by ϕ preserves ΘH^2 . After normalizing f and identifying ΘH^2 unitarily with H^2 , the compression has the block form

$$T_\phi^M = \begin{pmatrix} \gamma_f & 0 \\ g & M_\phi \end{pmatrix}$$

for some $g \in H^2$. Since ϕ is analytic past the closed disk, $M_\phi = \phi(M_z)$ and spectral mapping gives $\sigma(M_\phi) = \phi(\overline{\mathbb{D}})$.

For $\lambda \neq \gamma_f$, elementary block factorization shows that $T_\phi^M - \lambda$ is invertible exactly when $M_\phi - \lambda$ is invertible. At $\lambda = \gamma_f$, the lower-right block is invertible by hypothesis while the scalar diagonal block vanishes; hence the full operator is not invertible. This proves the displayed spectrum. Because $\phi(\overline{\mathbb{D}})$ is compact and does not contain γ_f , the singleton is an isolated connected component. Solving the lower block with $(M_\phi - \gamma_f)^{-1}$ also gives a nonzero eigenvector. $\square$

4 Neil-algebra counterexample

Consider the Neil representation space

$$M_N = \{h \in H^2 : h(0) = -h'(0)\} = \mathbb{C}f_N \oplus z^2H^2, \quad f_N(z) = \frac{1-z}{\sqrt{2}}.$$

This corresponds to the source parameters $(\alpha, \beta) = (2^{-1/2}, -2^{-1/2})$. Take

$$\phi(z) = \frac{z}{(1-z/2)^2}.$$

The symbol is bounded and analytic on a neighborhood of $\overline{\mathbb{D}}$. Since its constant and linear Taylor coefficients are 0 and 1, respectively,

$$\gamma_N = \langle \phi f_N, f_N \rangle = -\frac{1}{2}.$$

Moreover,

$$\phi(z) = -\frac{1}{2} \iff 2z = -(1-z/2)^2 \iff (z+2)^2 = 0.$$

The only preimage is $z = -2$, so $-1/2 \notin \phi(\overline{\mathbb{D}})$. Lemma 1 gives the exact disconnected spectrum

$$\sigma(T_\phi^{M_N}) = \left\{ -\frac{1}{2} \right\} \cup \phi(\overline{\mathbb{D}}).$$

5 Two-point counterexample

Now set

$$a = -\frac{1}{10}, \quad b = \frac{1}{10}, \quad t = \frac{11}{9},$$

and let $\Theta = B_{a,b}$. The function

$$f(z) = \frac{1-z}{1-z^2/100}$$

belongs to K_Θ . Direct evaluation gives

$$\frac{f(a)}{f(b)} = \frac{1+1/10}{1-1/10} = \frac{11}{9} = t.$$

Consequently the two-point Hardy space in the source has the orthogonal decomposition

$$H_t^2 = \mathbb{C}f \oplus B_{a,b}H^2.$$

We compute its scalar block exactly. Put $q = 1/100$. The Taylor coefficients of f occur in pairs,

$$f_{2j} = q^j, \quad f_{2j+1} = -q^j,$$

and $\|f\|^2 = 2/(1-q^2)$. Therefore the normalized correlations are

$$\frac{\langle z^{2\ell}f, f \rangle}{\|f\|^2} = q^\ell, \quad \frac{\langle z^{2\ell+1}f, f \rangle}{\|f\|^2} = -\frac{1+q}{2}q^\ell.$$

Since

$$\phi(z) = \sum_{k \geq 1} \frac{k}{2^{k-1}} z^k,$$

putting $x = q/4$ and summing the two geometric derivatives yields

$$\begin{aligned}\gamma_t &= 4 \sum_{\ell \geq 1} \ell x^\ell - \frac{1+q}{2} \sum_{\ell \geq 0} (2\ell+1)x^\ell \\ &= \frac{4x}{(1-x)^2} - \frac{(1+q)(1+x)}{2(1-x)^2} = -\frac{79402}{159201}.\end{aligned}$$

Write $s = 79402/159201$. Exact arithmetic gives

$$\frac{4}{9} < s < \frac{1}{2}.$$

The equation $\phi(z) = -s$ is

$$\frac{s}{4}z^2 + (1-s)z + s = 0.$$

Its two roots are real and negative. The smaller absolute value is

$$\frac{2((1-s) - \sqrt{1-2s})}{s},$$

which exceeds 1 exactly when $s > 4/9$. Hence neither root lies in $\overline{\mathbb{D}}$, and $\gamma_t \notin \phi(\overline{\mathbb{D}})$. Lemma 1 now gives a second exact disconnected spectrum:

$$\sigma(T_\phi^t) = \left\{ -\frac{79402}{159201} \right\} \cup \phi(\overline{\mathbb{D}}).$$

6 Essential spectra remain connected

Theorem 1. *Let $M \subset H^2$ be any closed finite-codimensional subspace and $\phi \in L^\infty(\mathbb{T})$. Then the Fredholm essential spectrum of the compression satisfies*

$$\sigma_e(T_\phi^M) = \sigma_e(T_\phi).$$

Consequently every associated compression in Question 6.1, and more generally every finite-codimensional Hardy compression, has connected essential spectrum.

Proof. Let $F = H^2 \ominus M$, which is finite-dimensional, and extend the compression to $H^2 = M \oplus F$ by

$$\tilde{T} = T_\phi^M \oplus 0_F.$$

For $m \in M$ and $f \in F$,

$$(T_\phi - \tilde{T})(m + f) = T_\phi f + (I - P_M)T_\phi m.$$

The first term ranges in the finite-dimensional space $T_\phi F$ and the second in F . Thus $T_\phi - \tilde{T}$ has finite rank, and Atkinson's theorem gives

$$\sigma_e(\tilde{T}) = \sigma_e(T_\phi).$$

Adding or deleting the finite-dimensional zero summand does not affect Fredholmness (including at $\lambda = 0$), so $\sigma_e(T) = \sigma_e(T_\phi^M)$. The classical essential spectrum on the right is connected by the Widom–Douglas theorem quoted in the source's introduction. $\square$

7 Verification, novelty, and remaining scope

The included exact-arithmetic script checks the Neil factorization, the two-point correlation sum, the rational value $-79402/159201$, both strict bounds on s , both preimages outside the unit disk, $t = 11/9$, and $\|f\|^2 = 20000/9999$. No computation is used as a proof.

A bounded novelty search on 17 August 2026 covered the run's four lightweight indexes and web searches using the exact title, arXiv id, Question 6.1, the phrase *disconnected spectrum*, *Neil algebra*, and finite-codimensional Toeplitz compressions. The exact hits were the source and its published version; no later explicit resolution or matching counterexample was found. Novelty is plausible but not certified.

These counterexamples completely answer the ordinary-spectrum part of Question 6.1 in the negative for both named settings. Theorem 1 completely answers its essential-spectrum part in the positive, in greater generality. The relative point-spectrum part of Question 6.1 and Question 6.2 remain open here. Human review should focus on the block decomposition and the exact two-point correlation calculation. Verdict: **candidate counterexamples and full essential-spectrum theorem, likely valid.**

References

- [1] C. Felder, D. T. Pfeffer, and B. P. Russo, *Spectra for Toeplitz Operators Associated with a Constrained Subalgebra*, arXiv:2110.06379; Integral Equations Operator Theory 94 (2022), Article 15.